MODULE 5 PERIPHERAL NERVOUS SYSTEM

The Peripheral Nervous System.

THE PERIPHERAL NERVOUS SYSTEM

A reference for the peripheral nervous system video and lab. This page covers the structure of a nerve, the spinal nerves and their roots, the rami, the nerve plexuses, and the classification of sensory receptors. The cranial nerves are covered on their own sheet. The focus is on the structures and the job each one does.

BY THE END

1. Describe the structure of a nerve and name its connective tissue wrappings.
2. Identify the spinal nerves, their roots, and their rami.
3. Classify the sensory receptors by stimulus, location, and structure.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The peripheral nervous system

drsrennie-stack.github.io/new-build-bio4-solano/pns-concept-videos.html

The Peripheral Nervous System, an Overview

The peripheral nervous system is everything outside the brain and spinal cord. It carries input from the body's receptors in to the CNS, and carries the response back out to the effectors.

THE COMPONENTS OF THE SYSTEM

Structure	What it is
Peripheral nervous system	All neural structures outside the brain and spinal cord
Function	To carry sensory input from receptors to the CNS, and to carry motor output from the CNS to the effectors
Sensory (afferent) receptors	Structures that detect changes in the internal and external environment
Peripheral nerves	Bundles of axons that connect the CNS to the body
Ganglia	Clusters of neuron cell bodies located outside the CNS
Motor (efferent) endings	The terminals where motor neurons meet the muscles and glands
Cranial nerves	The twelve pairs of nerves that arise from the brain, covered on their own sheet
Spinal nerves	The thirty one pairs of nerves that arise from the spinal cord

The Structure of a Nerve

A nerve is a cordlike organ: many axons bundled together and wrapped in three layers of connective tissue.

THE NERVE AND ITS WRAPPINGS

Structure	What it is
Nerve	A cordlike organ of the PNS; a bundle of peripheral axons, some myelinated and some not, enclosed by connective tissue
Axon	A single nerve fiber; many axons make up a nerve
Fascicle	A bundle of axons grouped together
Endoneurium	The innermost wrapping, around each individual axon
Perineurium	The middle wrapping, the thicker layer around each fascicle
Epineurium	The outermost wrapping, around the entire nerve

HOLD ONTO THIS

The three wrappings nest from the inside out, and each one names the level it surrounds: endoneurium around one axon, perineurium around one fascicle, epineurium around the whole nerve.

The Spinal Nerves

Thirty one pairs of spinal nerves connect the cord to the body. Every one is a mixed nerve, carrying both sensory and motor axons.

THE SPINAL NERVE, ITS ROOTS, AND ITS EXIT

Structure	What it is
Spinal nerves	The thirty one pairs of nerves that arise from the spinal cord
Mixed nerves	Every spinal nerve carries both sensory and motor axons
Posterior (dorsal) root	The sensory root that brings axons into the cord
Anterior (ventral) root	The motor root that carries axons out of the cord
Intervertebral foramen	The opening between adjacent vertebrae where a spinal nerve exits the vertebral canal

The thirty one pairs are grouped by the region of the cord they arise from. Compare the count in each region.

THE SPINAL NERVES BY REGION

Region	Number of pairs
Cervical	Eight pairs
Thoracic	Twelve pairs
Lumbar	Five pairs
Sacral	Five pairs
Coccygeal	One pair

HOLD ONTO THIS

The posterior root brings signals in and the anterior root carries them out. Posterior is sensory, anterior is motor, and the two join just past the cord to make one mixed spinal nerve.

The Rami of a Spinal Nerve

Just past the intervertebral foramen, each spinal nerve splits into branches called rami. Each ramus serves a different territory. Compare them.

THE RAMI AND WHAT EACH ONE SUPPLIES

Ramus or branch	What it supplies
Posterior (dorsal) ramus	The deep muscles and the skin of the posterior surface of the trunk
Anterior (ventral) ramus	The muscles and skin of the limbs and the lateral and anterior trunk; these rami form the plexuses
Meningeal branch	Re-enters the vertebral canal to supply the vertebrae, the ligaments, the blood vessels of the cord, and the meninges
Rami communicantes	Small branches off the anterior ramus that carry autonomic fibers to and from the sympathetic trunk

The Nerve Plexuses

The anterior rami of most spinal nerves do not run straight to their targets. They first join into networks called plexuses. The thoracic rami are the exception.

THE PLEXUSES AND THE THORACIC EXCEPTION

Structure	What it is
Nerve plexus	A network formed by the anterior rami of several spinal nerves
Cervical plexus	Formed from C1 to C4; serves the neck and shoulders, and gives off the phrenic nerve to the diaphragm
Brachial plexus	Formed from C5 to T1; serves the shoulder and the upper limb
Lumbar plexus	Formed from L1 to L4; serves the lower abdominal wall and part of the lower limb
Sacral plexus	Formed from L4 to S4; serves the buttock, the perineum, and most of the lower limb
Intercostal nerves	The anterior rami of T2 to T12, which do not form a plexus but run straight to the thoracic wall

The plexuses are covered in more detail, including the region affected by damage, on the Nerve Plexuses sheet.

Classifying Sensory Receptors

Sensory receptors are classified in three ways: by the type of stimulus they detect, by where they are located, and by their structure.

By stimulus type

WHAT KIND OF CHANGE THE RECEPTOR ANSWERS TO

Structure	What it is
Mechanoreceptors	Respond to mechanical force such as touch, pressure, vibration, and stretch
Thermoreceptors	Respond to changes in temperature
Photoreceptors	Respond to light; found in the retina of the eye
Chemoreceptors	Respond to chemicals in solution, such as smell and taste molecules
Nociceptors	Respond to potentially damaging stimuli and produce the sensation of pain

By location

WHERE THE RECEPTOR SITS

Structure	What it is
Exteroceptors	Detect stimuli at or near the body surface, such as touch, temperature, and the special senses
Interoceptors	Also called visceroceptors; detect stimuli inside the body, in the organs and blood vessels
Proprioceptors	Detect the position and movement of the body, in muscles, tendons, and joints

By structure

HOW THE NERVE ENDING IS BUILT

Structure	What it is
Free nerve endings	Nonencapsulated receptors; bare dendrite tips found in most tissues
Encapsulated receptors	Nerve endings wrapped in a connective tissue capsule

The Sensory Receptors

The named sensory receptors, with where each one is found and what it detects. Compare them.

THE NAMED RECEPTORS, THEIR LOCATION, AND THEIR STIMULUS

Receptor	Location	What it detects
Free nerve endings	Most body tissues, dense in connective tissue and epithelium	Pain, temperature, and crude touch
Tactile (Merkel) discs	The deepest layer of the epidermis	Light, discriminative touch
Hair follicle receptors	Wrapped around hair follicles	Movement of the hair
Tactile (Meissner) corpuscles	The dermal papillae of hairless skin	Fine, discriminative touch
Lamellar (Pacinian) corpuscles	The deep dermis and the hypodermis	Deep pressure and vibration
Bulbous (Ruffini) corpuscles	The deep dermis and joint capsules	Continuous, steady pressure
Muscle spindles	Within skeletal muscles	The stretch of a muscle
Tendon organs	Within tendons	The tension a muscle places on its tendon
Joint kinesthetic receptors	The capsules of synovial joints	Joint position and movement

FOLLOWING THE NERVE TO THE DEFICIT

Because a nerve is wrapped in connective tissue, it has some protection and some ability to heal. The **epineurium** and **perineurium** hold a cut nerve's shape, which is what lets a surgeon line up the fascicles and repair it.

The rami explain referred patterns. The **anterior rami** carry most limb signals, so a problem in the anterior rami or the plexus they form affects the limbs, while a **posterior ramus** problem stays in the muscles and skin of the back.

Receptors are not spread evenly. The fingertips and lips are crowded with tactile corpuscles and Merkel discs, which is why those areas feel fine detail so well, while the skin of the back, with far fewer, senses touch only crudely.